Heavy-Tailed Gradient Noise and Levy-Driven SGD

Simsekli, Sagun, and Gurbuzbalaban (2019). A Tail-Index Analysis of Stochastic Gradient Noise in Deep Neural Networks. arXiv:1901.06053v1.

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SGD

$$f(w) = \frac{1}{n}\sum_{i=1}^{n} f^{(i)}(w), \qquad w^\star = \arg\min_{w\in\mathbb{R}^p} f(w).$$

SGD uses a minibatch Ω_k of size b:

$$w_{k+1} = w_k - \eta\nabla \tilde{f}_k(w_k), \qquad \nabla \tilde{f}_k(w) = \frac{1}{b}\sum_{i\in\Omega_k} \nabla f^{(i)}(w).$$

Define stochastic gradient noise

$$U_k(w) = \nabla \tilde{f}_k(w) - \nabla f(w).$$

Brownian Model

The standard modeling assumption is

$$U_k(w) \sim \mathcal{N}(0, \sigma^2 I).$$

Then the SGD update can be written as

$$w_{k+1} = w_k - \eta \nabla f(w_k) + \sqrt{\eta}\sqrt{\eta\sigma^2}\, Z_k, \qquad Z_k \sim \mathcal{N}(0, I).$$

For small η, this suggests the SDE

$$dw_t = -\nabla f(w_t)\, dt + \sqrt{\eta\sigma^2}\, dB_t.$$

Why the Paper Challenges This Baseline

The Gaussian/Brownian picture has two tensions.

1. Empirical tension: measured gradient noise in deep nets has much heavier tails than Gaussian noise.
2. Theoretical tension: Brownian paths are continuous, so escaping a basin usually means climbing the barrier.

For Brownian-driven small-noise dynamics, first exit times scale roughly like

$$\exp\left(\frac{2H}{\varepsilon^2}\right),$$

where H is a relevant barrier height.

That makes depth more important than width, which clashes with the folklore that SGD tends to prefer wide minima.

Gradient Noise Is Not Gaussian

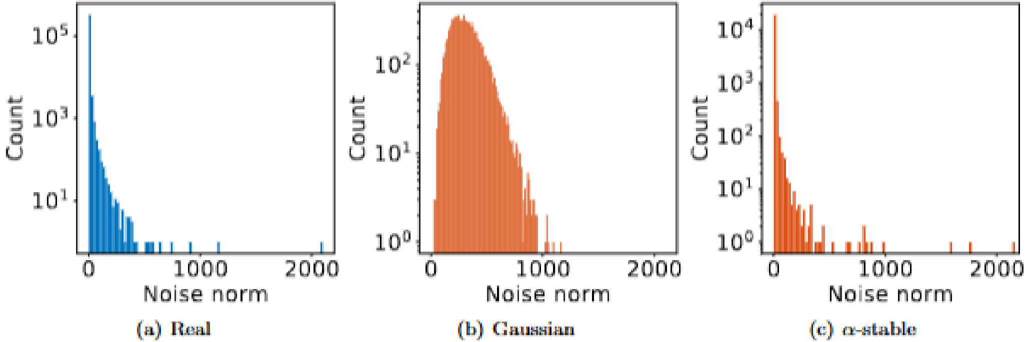


Figure 1: Comparison of Gradient Noise Norms

Stable Laws

The generalized central limit theorem says:

- finite variance sums $\Rightarrow$ Gaussian limit;
- heavy-tailed sums with infinite variance $\Rightarrow$ α-stable limit.

The paper focuses on symmetric α-stable random variables:

$$X \sim S\alpha S(\sigma) \quad \Longleftrightarrow \quad \mathbb{E}e^{i\omega X} = \exp(-|\sigma\omega|^\alpha).$$

Key facts:

- $\alpha \in (0, 2]$ is the tail index;
- smaller α means heavier tails;
- $\mathbb{E}|X|^r < \infty$ if and only if $r < \alpha$;
- $\alpha = 2$ recovers a Gaussian: $S2S(\sigma) = \mathcal{N}(0, 2\sigma^2)$.

From Stable Noise to Levy-Driven SDE

The paper assumes each coordinate of the gradient noise is stable:

$$[U_k(w)]_i \sim S\alpha S(\sigma(w)).$$

Then SGD can be rewritten as

$$w_{k+1} = w_k - \eta \nabla f(w_k) + \eta^{1/\alpha} \left(\eta^{(\alpha-1)/\alpha} \sigma\right) S_k,$$

where each coordinate of S_k has law $S\alpha S(1)$.

For small η, the continuous-time model becomes

$$dw_t = -\nabla f(w_t)\, dt + \eta^{(\alpha-1)/\alpha} \sigma\, dL_t^{\alpha}.$$

Here L_t^{α} is an α-stable Levy motion.

Definition of α-Stable Levy Motion

In the vector SDE above, each coordinate of L_t^α is an independent scalar α-stable Levy motion on $\mathbb{R}$. For $\alpha \in (0, 2]$, such a scalar process is determined by:

- **(i)** $L_0^\alpha = 0$ almost surely.
- **(ii)** For $t_0 < t_1 < \cdots < t_N$, the increments $L_{t_i}^\alpha - L_{t_{i-1}}^\alpha$ are independent $(i = 1, \ldots, N)$.
- **(iii)** For $s < t$, the increment $L_t^\alpha - L_s^\alpha$ has the same law as L_{t-s}^α, namely $S\alpha S((t-s)^{1/\alpha})$.
- **(iv)** Stochastic continuity: for every $\delta > 0$ and $s \geq 0$, $\mathbb{P}(|L_t^\alpha - L_s^\alpha| > \delta) \to 0$ as $t \to s$.

When $\alpha = 2$, L_t^α agrees with $\sqrt{2}\, B_t$ for standard Brownian motion B_t.

Brownian Motion vs. Levy Motion

Feature	Brownian motion B_t	Stable Levy motion L_t^α
Increment law	Gaussian	α-stable
Tail behavior	light-tailed	heavy-tailed for $\alpha < 2$
Variance	finite	infinite for $\alpha < 2$
Paths	almost surely continuous	can have jumps
Basin escape	climb out	jump out

The crucial behavioral change is discontinuity: large jumps make barrier height less controlling. $\alpha = 2$ gives scaled Brownian motion.

Stable Densities and Levy Paths

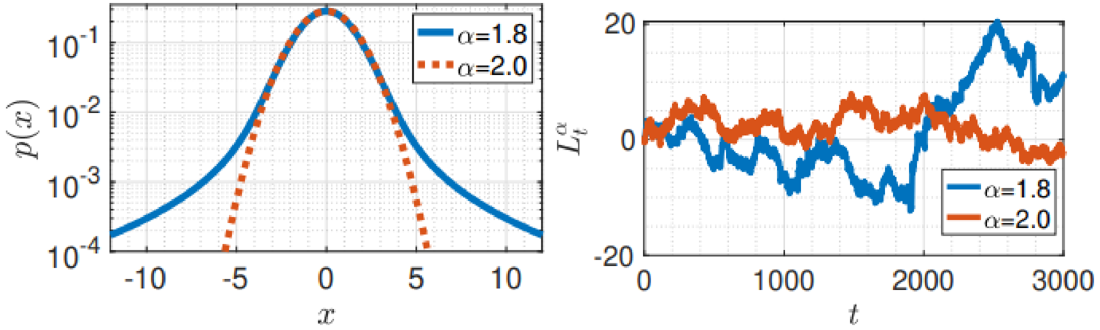


Figure 2: SαS Densities and Lévy Paths

Figure 2: Left - SαS densities; right - sample paths of L_t^α for $p = 1$. For $\alpha < 2$, the density has heavier tails and L_t^α displays jumps.

Metastability Setup

To isolate the basin-escape mechanism, the paper studies the 1D small-noise SDE:

$$dw_t^{\varepsilon} = -\nabla f(w_t^{\varepsilon})\, dt + \varepsilon\, dL_t^{\alpha}.$$

Assume f has local minima $m_1, \dots, m_r$ separated by local maxima

$$-\infty = s_0 < m_1 < s_1 < \cdots < s_{r-1} < m_r < s_r = \infty.$$

Each valley:

$$S_i = (s_{i-1}, s_i), \qquad L_i = |s_i - s_{i-1}|.$$

Question: Starting near m_i, how long until the process reaches another basin?

Two Valleys

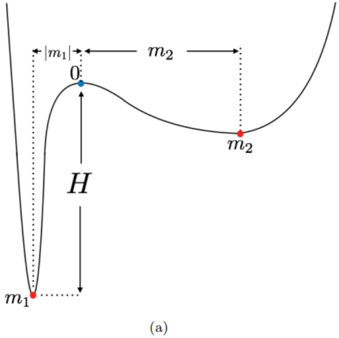


Figure 3: An objective with two local minima m_1, m_2 separated by a local maximum at $s_1 = 0$.

Theorem 1: Levy Exit Times

For w_0 near m_i, as $\varepsilon \to 0$, transition times satisfy

$$\mathbb{P}_{w_0}(\varepsilon^\alpha T^i(\varepsilon) \geq u) \leq e^{-q_i u}.$$

The transition probabilities are controlled by

$$q_{ij} = \frac{1}{\alpha}\left|\frac{1}{|s_{j-1} - m_i|^\alpha} - \frac{1}{|s_j - m_i|^\alpha}\right|, \qquad q_i = \sum_{j \neq i} q_{ij}.$$

Interpretation:

- transition time is polynomial in noise scale, on the order of $\varepsilon^{-\alpha}$;
- the rates depend on distances to valley boundaries;
- basin height does not appear.

Theorem 2: A Markov Chain Over Minima

After rescaling time by $\varepsilon^{-\alpha}$, the Levy-driven process converges to a continuous-time Markov chain on the minima:

$$w^{\varepsilon}_{t\varepsilon^{-\alpha}} \Rightarrow Y_{m_i}(t), \qquad Y(t) \in \{m_1, \ldots, m_r\}.$$

The generator has off-diagonal entries

$$q_{ij} = \frac{1}{\alpha} \left| \frac{1}{|s_{j-1} - m_i|^{\alpha}} - \frac{1}{|s_j - m_i|^{\alpha}} \right|,$$

and diagonal entries

$$q_{ii} = -\sum_{j \neq i} q_{ij}.$$

The stationary probabilities solve

$$Q^T \pi = 0.$$

Wide Valleys in the Two-Minima Case

For two minima with $m_1 < 0 < m_2$ and the barrier at $s_1 = 0$, the paper gives

$$\pi_1 = \frac{|m_1|^\alpha}{|m_1|^\alpha + |m_2|^\alpha}, \qquad \pi_2 = \frac{|m_2|^\alpha}{|m_1|^\alpha + |m_2|^\alpha}.$$

So if $|m_2| > |m_1|$, the second valley is wider and

$$\pi_2 > \pi_1, \qquad \frac{\pi_2}{\pi_1} = \left(\frac{|m_2|}{|m_1|}\right)^\alpha.$$

Estimating the Tail Index

The paper uses an estimator specialized for stable laws.

Let $X_1, \dots, X_K \sim S\alpha S(\sigma)$ and $K = K_1 K_2$. Define block sums

$$Y_i = \sum_{j=1}^{K_1} X_{j+(i-1)K_1}, \qquad i = 1, \dots, K_2.$$

The estimator is

$$\widehat{\frac{1}{\alpha}} = \frac{1}{\log K_1} \left(\frac{1}{K_2} \sum_{i=1}^{K_2} \log |Y_i| - \frac{1}{K} \sum_{i=1}^{K} \log |X_i| \right).$$

Then $\widehat{1/\alpha} \to 1/\alpha$ almost surely as $K_2 \to \infty$.

Tail Estimator

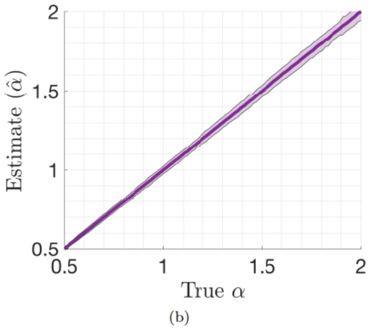


Figure 4: Tail-index estimator illustration.

Experimental Protocol

At selected SGD iterations, the authors:

1. compute the full gradient $\nabla f(w_k)$;
2. partition the data into disjoint minibatches of size b;
3. compute minibatch gradients $\nabla \tilde{f}_{\Omega_k^i}(w_k)$;
4. form gradient noises

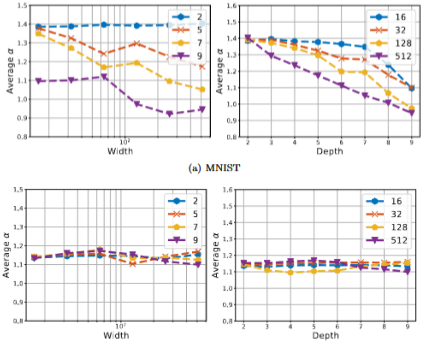
$$U_k^i(w_k) = \nabla \tilde{f}_{\Omega_k^i}(w_k) - \nabla f(w_k);$$

5. concatenate coordinates and estimate α.

Experiments vary:

- architecture: fully connected nets and AlexNet-like CNNs;
- dataset: MNIST, CIFAR10, CIFAR100;
- network size, minibatch size, loss function, and random seed.

Results: Network Size - FCN Width and Depth



(a) MNIST

(b) CIFAR10

Figure 5: Estimation of α for varying widths and depths in FCNs. The curves in the left figures correspond to different depths, and the ones on the right figures correspond to widths.

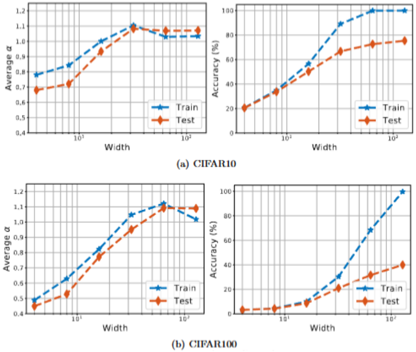
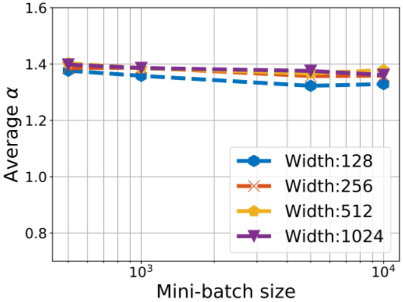
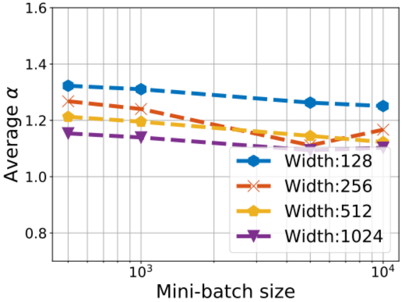


Figure 6: The accuracy and $\hat{\alpha}$ of the CNN for varying widths.

Results: Minibatch Size



(a) Depth = 2



(b) Depth = 4

Figure 7: Estimation of α for varying minibatch size.

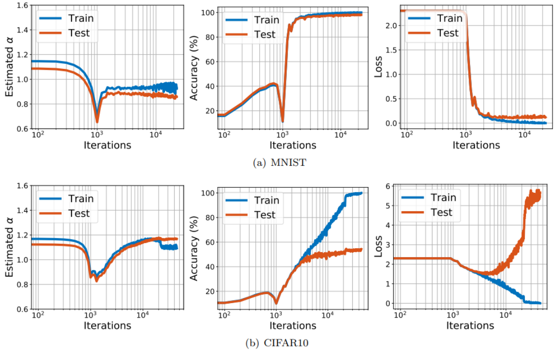


Figure 8: The iteration-wise behavior of α for the FCN.

Interpretation of the Results

- early phase: α rapidly decreases;
- near minimum α: the SGD trajectory shows a jump-like event;
- later phase: α stabilizes while accuracy improves.

The authors interpret this as evidence that early SGD dynamics may include barrier crossing or large basin transitions.

They do not prove whether the jump lands in a different basin or a better part of the same basin.

Brownian Results vs. Levy Results

Question	Brownian model	Levy model
Noise law	Gaussian	α-stable
CLT justification	finite variance CLT	generalized CLT
Continuous-time driver	B_t	L_t^α
Path geometry	continuous	jumps for $\alpha < 2$
Escape scaling	exponential in barrier height	polynomial in ε^{-1}
What controls preference?	depth/barrier height dominates	width appears directly
Wide minima story	hard to explain locally	supported by metastability

What Is Proven vs. What Is Inferred

Proven or cited mathematical facts:

- stable laws arise from a generalized CLT;
- stable Levy motion has α-stable increments;
- Levy-driven small-noise dynamics has width-dependent transition rates;
- in simple cases, stationary probabilities favor wider valleys.

Empirical evidence:

- measured gradient noise in deep nets is heavy-tailed;
- estimated α is consistently below 2;
- minibatch size does not restore Gaussianity in their experiments.

Inference:

If SGD gradient noise is well modeled as α-stable, then existing Levy metastability theory gives a plausible mechanism for SGD's preference for wide minima.

Limitations and Open Problems

- The dependence on learning rate and minibatch size is not fully characterized.
- The tail index may change with the current state w_k and time.
- Deep-network landscapes have many nearly flat directions, while the clean theory assumes nondegenerate minima.
- Empirical heavy tails do not by themselves prove that an exact stable law governs every coordinate.

Closing Summary

heavy-tailed gradient noise $\Rightarrow$ α-stable model $\Rightarrow$ Levy-driven SDE $\Rightarrow$ jumps $\Rightarrow$ width-dependent